Context-Based Note Taking

The Concept

A handwriting-first note-taking app where uploaded course materials (PDFs, readings, etc.) are automatically retrieved and can be cited inline as you write. During a lecture, when you jot down something that connects to your readings the app uses RAG to surface the relevant passage from your uploaded content instantly, so you can cite it without ever leaving your notes.

Tech Stack & Platform

Frontend / Backend
Next.js / Fastify

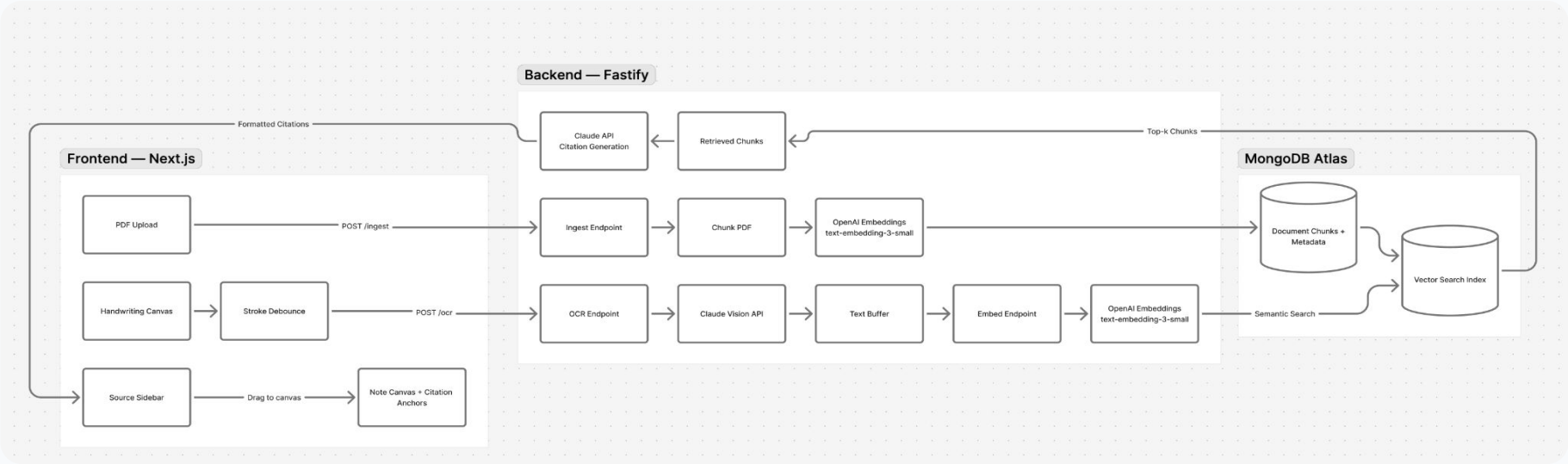
Data & Search
MongoDB Atlas + Vector

AI Engine
OpenAI API

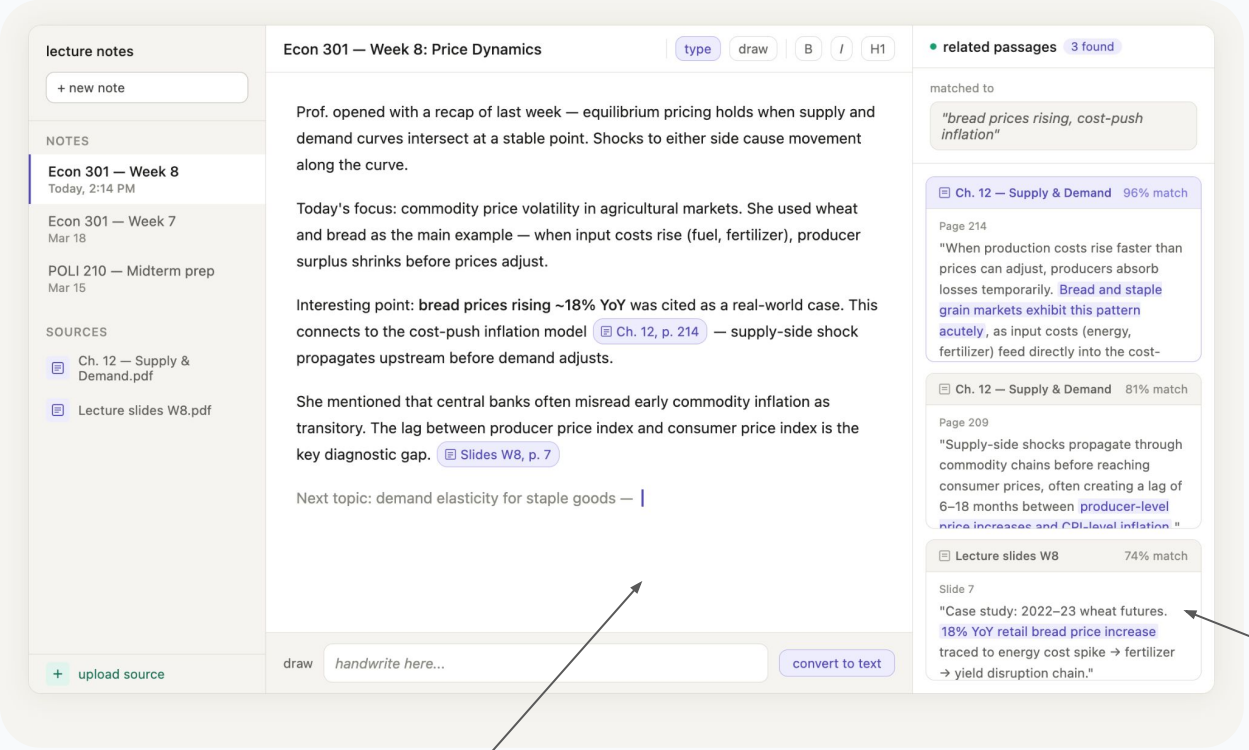
OCR Implementation
Claude Vision API

Preliminary Work

Developed a comprehensive technical flowchart mapping the RAG-enabled note-taking pipeline



Design Mock



Accompanied by AI provided relevance scores and summaries

This will be a handwritable canvas

Will be able to drag and drop citations

Initial Prompt

i want to make an app for note taking. Heres the flow
* you upload content related to the lecture or whatever. Could be a pdf of a reading or something
* Then during the lecture, you take notes by handwriting or typing
* Then lets say the prof mentions something about bread rising in price, and you recognize that part from the reading, and you want to connect them / cite in the notes
- Before you'd ctrl f the readings for bread and take a bunch of time
- but in this app it will automatically retrieve from the uploaded content via a RAG, allowing you to easily cite inline.

Claude validated the idea, emphasizing that the novelty is given by the real time suggestions aspect and handwriting first approach

But here's what's genuinely different about your framing:

All of those tools treat RAG as a *Q&A interface* — you finish your work, then go ask the AI questions. Your idea is different: **RAG as a real-time writing aid during the lecture itself.** The citation surfaces *inline, as you're composing notes*, not after the fact in a chat box. That's closer to how GitHub Copilot feels vs. how ChatGPT feels — ambient and contextual rather than interrupt-driven.

The closest thing I found to this is Anara, which links AI responses to source passages with "verifiable source highlighting" Anara — but it's still fundamentally a post-hoc Q&A tool, not an inline writing companion.

What would actually make this stand out:

1. **The trigger mechanic** — if it auto-surfaces suggestions *while you type* (not just when you ask), that's meaningfully new UX in this space
2. **Inline citation as a first-class primitive** — not a chatbot response, but something that lives in the note itself like a footnote or chip
3. **Lecture-specific design** — upload readings *before* class, take notes *during*, connections get made *in the moment*. That temporal flow isn't something NotebookLM or Reor are designed around